MATHEMATICS FULL TEST (PREDICTED SET 3)

Expert Analysis Series • Boards Dominator by Aashi

Time: 3 Hours | Maximum Marks: 80 | Total Questions: 33

SECTION A — MCQs (1 × 12 = 12 Marks)

Q1. If A is a square matrix and $|A|=3$, then $|A.adj\ A|$ equals:

(a) 3 (b) 6
(c) 12 (d) 24

Q2. If $f(x) = |x|$, then at $x = 0$, the function is:

(a) Continuous & differentiable (b) Continuous but not differentiable
(c) Differentiable only (d) Neither

Q3. Value of $\int_{-1}^{1} x^3\, dx$ is:

(a) 0 (b) 1
(c) 2 (d) -1

Q4. Integrating factor of $dy/dx + 2y = x$ is:

(a) e^x (b) e^{2x}
(c) 2x (d) e^{-2x}

Q5. If vectors are perpendicular, then their dot product is:

(a) 1 (b) -1
(c) 0 (d) Undefined

Q6. Maximum value of $\sin x$ is:

(a) 1 (b) 5
(c) No maximum (d) 4

Q7. If events A and B are independent, then $P(A \cap B) = ?$

(a) P(A) + P(B) (b) P(A).P(B)
(c) 1 (d) 0

Q8. The order of the differential equation $d^2y/dx^2 + y = 0$ is:

(a) 1
(b) 2
(c) 3
(d) 0

Q9. Domain of $\sin^{-1} x$ is:

(a) R
(b) (-1,1)
(c) [-1,1]
(d) (0,1)

Q10. If the determinant of a matrix is zero, the matrix is:

(a) Identity
(b) Singular
(c) Diagonal
(d) Symmetric

Q11. The slope of the tangent at maxima is:

(a) 1
(b) -1
(c) 0
(d) Undefined

Q12. If $|a|=3$, $|b|=4$, and the angle between them is $90°$, then $|a+b|$ is:

(a) 5
(b) 7
(c) 1
(d) 12

SECTION B — Very Short Answer (2 × 6 = 12 Marks)

Q13. Differentiate $\sin(x^2)$ w.r.t. x.

Q14. Find the area of a parallelogram whose diagonals are $d_1 = 3i+j-2k$ and $d_2) = i-3j+4k$.

Q15. Evaluate: $\tan^{-1}(1) + \tan^{-1}(1)$.

Q16. Find the equation of the tangent to $y = x^2$ at $(1,1)$.

Q17. Find the mean if the probability distribution is: X: [0, 1, 2], $P(X)$: [1/4, 1/2, 1/4].

Q18. Solve: $\int e^x dx$.

SECTION C — Short Answer (3 × 8 = 24 Marks)

Q19. Using matrices, solve the system: $x+y=5, x-y=1$.

Q20. Find the intervals in which $f(x) = x^2 - 4x + 6$ is strictly increasing/decreasing.

Q21. Evaluate: $\int [x / (1+x^2)]\, dx$.

Q22. Find the angle between vectors $a = i+j-k$ and $b = i-j+k$.

Q23. Solve the differential equation: $dy/dx = x+y$.

Q24. Linear Programming Problem: Maximise $Z = x + y$ Subject to: $x+y \le 4, x,y \ge 0$. Find the optimal value.

Q25. Find the probability of getting exactly one head when tossing two coins.

Q26. Verify Mean Value Theorem for $f(x) = x^2$ on $[1,3]$.

SECTION D — Long Answer (5 × 3 = 15 Marks)

Q27. Evaluate: $\int [(x^2+1) / (x^2+2)]\, dx$.

Q28. Find the shortest distance between lines:
$r = (i+j) + \lambda(i-k)$
$r = (2i-j) + \mu(j+k)$

Q29. Find the absolute maximum and minimum of $f(x) = x^3$ on $[1,5]$.

SECTION E — Case Study (4 × 2 = 8 Marks)

A solar panel installation forms a rectangle of area A.
Q30. Write the expression for Area.
Q31. If the perimeter P is fixed, express A in terms of one dimension x.
Q32. Find the value of x for which the area is maximum.
Q33. State the maximum area obtained when $P=40$.

⭐ WHY THIS PAPER IS HIGHLY ACCURATE

Based on the analysis of Series EF1GH/5, PQ1RS/1, and current board trends, these chapters appeared in EVERY set:

- Matrices & Determinants (System solving focus)
- Continuity & Differentiability (Chain rule focus)
- Application of Derivatives (Optimisation focus)
- Integrals (Substitution & Properties focus)
- Differential Equations (Linear & Homogeneous)
- Vectors & 3D Geometry (Shortest Distance focus)
- Linear Programming (Graphical focus)
- Probability (Bayes' & Distribution focus)

Mathematics Mock Test Set 3 By Aashi • Thunderstudy
Solve Each Question with Precision.